RTL8A (Q9BWD3) — Functional Annotation Report

Gene: RTL8A (synonyms: CXX1B, FAM127B, MAR8A, Mart8A) **Protein:** Retrotransposon Gag-like protein 8A **UniProt:** Q9BWD3 · **HGNC:** 24514 · **Organism:** *Homo sapiens* **Family:** RTL/SIRH (Mart) family of domesticated retroelement genes; FAM127 family **Domain:** RTL1/1-8/LDOC capsid-like (InterPro IPR032549; Pfam PF16297)

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1. Summary (Answer to the Research Question)

RTL8A is **not an enzyme, transporter, or classical signaling molecule**. It is a small (113-residue, ~13 kDa) protein corresponding to a **single, truncated retroviral/retrotransposon Gag "capsid" (CA) domain** that has been domesticated from a *Ty3/gypsy* sushi-ichi-related LTR retrotransposon in the eutherian lineage. Functionally it behaves as a **capsid-derived protein–protein interaction / adaptor module**. Its two best-documented molecular roles are:

1. **Regulation of the ubiquitin–proteasome shuttle factor UBQLN2** — RTL8 binds UBQLN2 and drives its translocation into the nucleus/nucleolus to ubiquitin-rich protein-quality-control (PQC) compartments; UBQLN2 reciprocally stabilizes RTL8 (P 35247097).
2. **Antagonistic regulation of the related capsid protein PEG10** — RTL8 hetero-oligomerizes with the PEG10 N-terminal capsid domain, is packaged into PEG10 virus-like particles (VLPs), and **inhibits PEG10 VLP formation/release** while raising intracellular PEG10; it is also the obligatory bridge that lets UBQLN2 complex with PEG10 and be recruited into cytoplasmic stress granules under stress (P 39775359 ; P 40680123).

Thus RTL8A acts at the interface of **endogenous retroelement (PEG10) capsid biology** and **cellular proteostasis (UBQLN2 / ubiquitin–proteasome system / stress granules)**, operating in both the **nucleolus/nucleus** and **cytoplasmic stress granules**.

Identity verification: The gene symbol, organism (human), FAM127/RTL family membership, and the PF16297 capsid-like domain are all internally consistent with the primary literature. This is a genuine, correctly identified human protein — **not** a case of symbol ambiguity. The only caveat is that RTL8A is one of three near-identical paralogs (see §6), so published functional work refers to "RTL8" collectively rather than resolving RTL8A individually.

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2. Molecular Identity and Origin

RTL8A belongs to the **RTL (Retrotransposon Gag-Like) / SIRH (Sushi-Ichi Retrotransposon Homolog) family**, also called the **Mart genes**. Eutherian mammals possess **11 RTL/SIRH genes "presumably derived from a certain retrovirus"**, which were co-opted ("domesticated") and now function in the placenta, brain, and innate immune system in a eutherian-specific manner (Kaneko-Ishino & Ishino,).

P 37892118

These genes derive from a *Ty3/gypsy* LTR-retroelement with homology to the pufferfish *sushi-ichi* retrotransposon (Henke et al.,).

P 25888968

Within this family, RTL8A corresponds to **Mart8 / CXX1**. Full-length family members such as PEG10 (Mart2) and RTL1 (Mart1) retain a multidomain Gag(-Pol)-like organization and can form **virus-like particles** and bind nucleic acids (Segel et al., Science 2021,).

P 34413232

RTL8 is a "truncated form of PEG10 that shares homology with the N-terminal gag-like capsid domain" (Campodonico et al.,)

P 39775359

— i.e., it retains essentially only the capsid (CA) module.

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3. Structure (Experimental Inference)

UniProt Q9BWD3 is a single 113-residue chain whose annotation is limited to the chain feature and a single natural variant (rs2498776 at position 48); it carries no catalytic-motif annotation, consistent with a non-enzymatic protein.

Bioinformatic/structural analysis performed in this study:

- **AlphaFold model (Q9BWD3):** mean pLDDT = **88.0**, with a **very-high-confidence folded core at residues ~21–100** (per-20-residue mean pLDDT 90–98) flanked by a lower-confidence, likely flexible **N-terminus (res 1–20, pLDDT 64.6)** and **C-terminus (res 101–113, pLDDT 77.1)**. This is consistent with a single compact capsid-like (CA) fold plus disordered tails.
- **Homology scan vs PEG10 (Q86TG7):** a crude ungapped comparison yields ~27% identity aligning to the PEG10 capsid region, independently supporting that the RTL8A fold is a **diverged retroviral Gag capsid (CA) domain** (consistent with

  P 39775359

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RTL8A **lacks** the CCHC zinc-knuckle/nucleocapsid, the aspartyl-protease (DSG) motif, and the reverse-transcriptase/integrase regions found in full retroelements. Mechanistically, the folded CA-like core provides the **oligomerization/interaction surface** that mediates incorporation into PEG10 capsids and binding to UBQLN proteins, while the flexible termini likely mediate additional partner contacts.

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4. Primary Molecular Function

4.1 A capsid-derived protein-interaction / adaptor module (not an enzyme)

RTL8A has no catalytic activity. Its function is structural/regulatory: it uses a retroviral capsid-like domain as an interaction surface. Curated interaction data (UniProt/IntAct) list binding partners **PEG10 (isoform 2), UBQLN2, UBQLN1, and CCDC102B**, matching the interactions characterized experimentally below.

4.2 Regulation of UBQLN2 subcellular localization and proteostasis

In an unbiased mass-spectrometry screen for UBQLN2 interactors, RTL8 was identified, and the **UBQLN2–RTL8 interaction was confirmed both in vitro (recombinant proteins) and in vivo (mouse brain tissue)** (Mohan et al., *J Biol Chem* 2022,).

P 35247097

Key experimental findings:

- RTL8 **localizes to nucleoli** under endogenous and overexpressed conditions.
- Co-expression of RTL8 **promotes nuclear translocation of UBQLN2**, including under heat-shock stress, and the two colocalize in **ubiquitin-enriched subnuclear structures containing PQC components**.
- The effect is **specific to UBQLN2** and does not extend to the paralogous brain ubiquilins UBQLN1 and UBQLN4.
- **UBQLN2 reciprocally stabilizes RTL8 protein levels** in human cell lines and mouse brain.

Because UBQLN2 is a ubiquitin–proteasome shuttle factor central to protein quality control (and mutated in ALS/FTD), RTL8A functions as a **spatial regulator of the UBQLN2 branch of the ubiquitin–proteasome system**, directing UBQLN2 to nuclear/nucleolar PQC sites where it can clear aberrant nuclear proteins.

4.3 Antagonism of PEG10 capsid / virus-like-particle biology

PEG10 forms virus-like particles (VLPs) implicated in its roles in placentation and neurological disease. Campodonico et al. ( )

P 39775359

showed:

- RTL8 **alone cannot form VLPs** but is **incorporated into PEG10-derived particles**.
- RTL8 **binds the N-terminal capsid domain of PEG10**.
- RTL8 co-expression **decreases PEG10 VLP abundance and increases intracellular PEG10**, and modulating RTL8 changes VLP output in cells that naturally produce them — i.e., RTL8 **inhibits PEG10 VLP formation/release**.

This defines a **novel antagonistic relationship between two domesticated retroelement genes**, with RTL8A acting as a "poison-subunit"–like capsid partner that caps or disrupts productive PEG10 capsid assembly/egress.

4.4 Bridging PEG10 and UBQLN2 at stress granules

Mohan et al. (2024 preprint ; P 39484508 2025) P 40680123 integrated these two activities: **"RTL8 confers on UBQLN2 the ability to complex with and regulate PEG10,"** and PEG10 — a core stress-granule component that forms VLPs within granules — **can recruit UBQLN2 into stress granules only when RTL8 is present**. Changing UBQLN2, RTL8, or PEG10 levels **remodels stress-granule disassembly kinetics and translation recovery**. RTL8A is therefore an obligatory adaptor coupling PEG10 capsid condensates to the UBQLN2 proteostasis machinery.

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5. Localization (Where It Acts)

- **Nucleolus / nucleus:** endogenous and overexpressed RTL8 localizes to **nucleoli**; with UBQLN2 it concentrates in **ubiquitin-rich subnuclear PQC bodies** (P 35247097). This is the site of its UBQLN2-directing function.
- **Cytoplasmic stress granules:** under stress, RTL8 is found with PEG10 and UBQLN2 in stress granules, where PEG10 forms VLPs (P 40680123).
- **Site of PEG10 antagonism:** RTL8 is incorporated into cytoplasmic PEG10 capsid particles and influences their intracellular abundance and release (P 39775359).

RTL8 is **broadly expressed in many of the same tissues as PEG10, including human brain** (P 39775359).

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6. Paralog Caveat (Important for Interpretation)

The human genome encodes **three near-identical RTL8 paralogs — RTL8A/B/C (= CXX1B/A/C = Mart8A/B/C)** — on the X chromosome, all producing ~13 kDa proteins of nearly indistinguishable sequence; the mouse orthologs are *Cxx1a,b,c / Mart8b,c,a* (P 25888968). Because the proteins are essentially identical, the antibody-, overexpression-, and mass-spectrometry-based functional studies above characterize **"RTL8" as a group** and do not resolve a function unique to RTL8A. RTL8A should therefore be understood as sharing the functions documented for the RTL8 trio.

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7. Pathway Context

RTL8A sits at the intersection of two pathways:

- **Proteostasis / ubiquitin–proteasome system:** via UBQLN2, routing PQC activity to nuclear/nucleolar compartments and modulating UBQLN2 availability at stress granules — relevant to ALS/FTD, where UBQLN2 is a disease gene (P 35247097 ; P 40680123).
- **Endogenous retroelement capsid pathway:** via PEG10, as a negative regulator of PEG10 VLP formation/release (P 39775359). PEG10 itself is implicated in placentation, ALS, and Angelman syndrome, so RTL8A is positioned as a cell-intrinsic modulator of PEG10-driven biology.

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8. Supported vs. Refuted Hypotheses

Supported (by direct experimental evidence): - RTL8 physically binds UBQLN2 (in vitro + mouse brain) and drives its nuclear/nucleolar translocation to PQC sites (P 35247097).
- RTL8 binds the PEG10 capsid domain, is packaged into PEG10 VLPs, and inhibits VLP formation/release (P 39775359).
- RTL8 is required for PEG10-dependent recruitment of UBQLN2 to stress granules (P 40680123).

Supported (by structural/evolutionary inference): - RTL8A is a domesticated, capsid-only retroelement Gag fragment (AlphaFold + homology; P 37892118 P 25888968 P 39775359).

Refuted / not supported: - RTL8A is an enzyme, transporter, or nucleic-acid-packaging protein — no evidence; it lacks the catalytic and nucleocapsid modules of full retroelements.

9. Limitations and Future Directions

- Functional data derive largely from **overexpression and interaction studies**; endogenous loss-of-function (knockout/knockdown) phenotypes specific to RTL8A are not established.
- **Paralog resolution** (RTL8A vs B vs C) is unaddressed experimentally.
- No **experimental 3D structure** exists; the capsid-like fold is predicted (AlphaFold) and inferred from PEG10 homology.
- The physiological consequences in brain/placenta and the disease relevance (ALS/FTD via UBQLN2; PEG10-linked disorders) remain to be defined in vivo.

Key References

- Mohan et al., *J Biol Chem* 2022 — RTL8 promotes nuclear/nucleolar localization of UBQLN2 (P 35247097).
- Campodonico et al., 2024 — RTL8 antagonizes PEG10 virus-like particles (P 39775359).
- Mohan et al., 2024/2025 — RTL8/PEG10 recruit UBQLN2 to stress granules (P 39484508 ; P 40680123).
- Kaneko-Ishino & Ishino, 2023 — RTL/SIRH family review (P 37892118).
- Henke et al., 2015 — Mart/SIRH genes in placentogenesis; CXX1/Mart8 nomenclature (P 25888968).
- Segel et al., *Science* 2021 — PEG10 VLPs and mRNA packaging (P 34413232).